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Editor-in-Chief
Computers and Electrical Engineering

Dear Sir/Ma'am,

We are pleased to submit our manuscript titled **“Efficient reformulations of ReLU deep neural networks for surrogate modelling in power system optimisation”** for consideration in the *Computers and Electrical Engineering*.

Our research presents an optimisation-with-constraint-learning framework that integrates machine learning surrogate models directly into optimisation problems. We develop a new linear programming (LP) reformulation for ReLU deep neural networks, enabling scalable and accurate representation of nonlinear relationships within mixed optimisation settings. This framework is applied to an aggregator bidding problem for distributed energy resources, where prosumer price responsiveness is learned via a ReLU DNN and embedded as an LP reformulation within the aggregator's profit-maximisation model.

The key contributions of this work are as follows:

- We introduce a method to convexify ReLU DNNs, producing tight LP reformulations suitable for efficient integration into optimisation workflows.
- We apply this within an optimisation-with-constraint-learning framework by learning prosumer responsiveness through a convexified ReLU DNN and embedding it within an operational decision-making model for an energy aggregator.
- We conduct an extensive evaluation of the proposed reformulation, benchmarking it against established alternatives and demonstrating strong performance across varied optimisation settings and neural network architectures.

We believe this manuscript is well aligned with the journal's focus on the application of artificial intelligence to make engineering problems tractable, particularly in leveraging ReLU DNNs for optimisation in energy market contexts.

We confirm that the manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. All authors have approved the manuscript and consent to its submission to *Computers and Electrical Engineering*.

Thank you for considering our submission. We look forward to your feedback.

Sincerely,
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